

Part I: Short answer item

1. Evaluate

i. $\lim_{x \rightarrow 0} \frac{\sqrt{2+3x} - \sqrt{2-3x}}{x} = \frac{3\sqrt{2}}{2}$

ii. $\lim_{x \rightarrow 0^+} \frac{1/x}{e^{1/x^2}} = \frac{1}{e}$

2. Let $f(x) = x^2 \ln(\sin x) \cos x$, then $\frac{d}{dx} f(x) = 2x \ln(\sin x) \cos x + x \cos^2 x \csc x - x^2 \ln(\sin x) \sin x$

3. For the function $f(x) = 3x^4 - 4x^3$, determine the intervals where the graph is

a) Increasing $[1, \infty)$

b) Decreasing $(-\infty, 1]$

c) Concave upward $(-\infty, 0], [\frac{2}{3}, \infty)$

d) Concave downward $[0, \frac{2}{3}]$

Part II: Workout problem item

1. Sketch the graph of $f(x) = 3x^5 - 5x^3$ by showing all the necessary steps.

2. Evaluate $\lim_{x \rightarrow 0} \frac{\ln(\sin 2x)}{\ln(\sin 3x)}$

3. Show that $\lim_{x \rightarrow 2} (x^2 - 1) = 3$ using formal definition of limit.

$f(x) = 3x^5 - 5x^3$

$\mathbb{R}$

Intercept $y=0$

$0 = 3x^5 - 5x^3$

$3x^3(x^2 - 5)$

$x=0$ $x=\pm\sqrt{5}$

$(0,0)$

$(\sqrt{5},0)$

$(-\sqrt{5},0)$

Intercept $x=0$

$3(0)^5 - 5(0)^3$

$y=0$

$(0,0)$

function

Symmetric w.r.t

origin

$\lim_{x \rightarrow \infty} 3x^5 - 5x^3 = \infty$ Does not have H.A

Does not have V.A b/c Domain = $\mathbb{R}$

$\frac{15x^4}{5x^3} = 3x$

$\frac{12x^3}{3x^2} = 4x$

$\frac{9x^2}{2x} = \frac{9}{2}x$

$$2, \lim_{x \rightarrow 0} \frac{\ln(\sin 2x)}{\ln(\sin 3x)} = \left(\frac{0}{0} \right)$$

$$\lim_{x \rightarrow 0} \frac{\frac{1}{\sin 2x} \cdot \cos 2x \cdot 2}{\frac{1}{\sin 3x} \cdot \cos 3x \cdot 3} = \frac{2 \cos 2x}{\sin 2x} \cdot \frac{\sin 3x}{3 \cos 3x}$$

$$\lim_{x \rightarrow 0} = \frac{2 \cos 2x}{2 \sin x \cos x} \cdot \frac{\sin 3x}{3 \cos 3x}$$

$$\lim_{x \rightarrow 0} = \frac{2 \cos 2x}{(2 \cos x)(3 \cos 3x)} \cdot \frac{\sin 3x}{\sin x}$$

$$\lim_{x \rightarrow 0} = \frac{\cos 2x}{\cos x (3 \cos 3x)} \cdot 3$$

$$\lim_{x \rightarrow 0} = \frac{1 \cdot 3}{3} = \frac{3}{3} = 1$$

$$2 \ln(\sin 2x) \cos x + 2 \cos^2 x \cos x - 2 \ln(\sin x) \sin x$$

$$f(x) = 3x^5 - 5x^3$$
$$f'(x) = 15x^4 - 15x^2$$

$$15c^4 - 15c^2 = 0$$

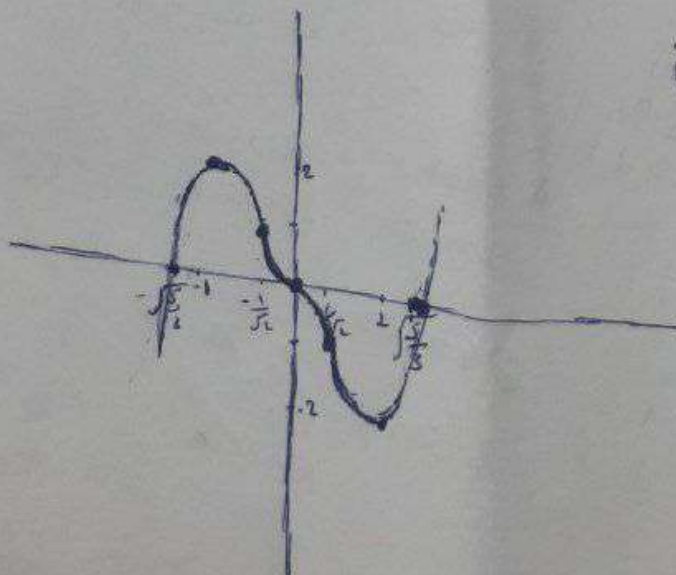
$C=0$ & $C=\pm 1$ are Critical no

$C=0$ & $C=\pm 1$ are Critical no

$-\infty, 1], [1, \infty)$ - increase

- $[-1, 1]$ - decrease

$x = 1$ maximum
relative



Part I: Short answer item

1. Evaluate

i. $\lim_{x \rightarrow 0} \frac{\sqrt{2+3x} - \sqrt{2-3x}}{x} = \frac{3\sqrt{2}}{2}$

ii. $\lim_{x \rightarrow 0^+} \frac{1/x}{e^{1/x^2}} = 1$

2. Let $f(x) = x^2 \ln(\sin x) \cos x$, then $\frac{d}{dx} f(x) = 2x \ln(\sin x) \cos x + x \cos^2 x \csc x - x^2 \ln(\sin x) \sin x$

3. For the function $f(x) = 3x^4 - 4x^3$, determine the intervals where the graph is

a) Increasing $[1, \infty)$

c) Concave upward $(-\infty, 0], [\frac{2}{3}, \infty)$

b) Decreasing $(-\infty, 1]$

d) Concave downward $[0, \frac{2}{3}]$

Part II: Workout problem item

1. Sketch the graph of $f(x) = 3x^5 - 5x^3$ by showing all the necessary steps.

2. Evaluate $\lim_{x \rightarrow 0} \frac{\ln(\sin 2x)}{\ln(\sin 3x)}$

3. Show that $\lim_{x \rightarrow 2} (x^2 - 1) = 3$ using formal definition of limit.

$f(x) = 3x^5 - 5x^3$

$\text{Domain} = \mathbb{R}$
 $\text{Range} = \mathbb{R}$

Intercept $y=0$

$0 = 3x^5 - 5x^3$

$x^3(x^2 - 5)$

$x=0 \quad x=\pm\sqrt{5}$

Intercept $x=0$

$3(0)^5 - 5(0)^3$

$y=0 \quad (0,0)$

function

Symmetric w.r.t

~~origin~~
origin

$\lim_{x \rightarrow \infty} 3x^5 - 5x^3 = \infty$

Does not have H.A

$\lim_{x \rightarrow -\infty}$

Does not have V.A b/c Domain = $\mathbb{R}$

$\frac{3x^5 - 5x^3}{\sin 2x}$

$\frac{15x^4 - 15x^2}{2 \cos 2x}$

$\frac{5x^4 - 5x^2}{2 \cos 2x}$

$\frac{20x^3 - 10x}{4 \cos 2x}$

$\frac{5x^3 - 5x}{2 \cos 2x}$

$\frac{15x^2 - 5}{4 \cos 2x}$

$\frac{15x^2 - 5}{4 \cos 2x}$

$$\lim_{x \rightarrow 2} (x^2 - 1) = 3$$

$$|f(x) - L| < \varepsilon \quad \text{whenever} \quad |x - 2| < \delta$$

$$|x^2 - 1 - 3| < \varepsilon$$

$$|x^2 - 4| < \varepsilon$$

$$|x^2 - 2| |x + 2| < \varepsilon$$

$$\text{let } \delta = 1$$

$$-1 < x - 2 < 1$$

$$1 < x < 3$$

$$3 < x + 2 < 5$$

$$|x - 2| |x + 2| < 5 |x - 2| < \varepsilon$$

$$|x - 2| < \frac{\varepsilon}{5}$$

$$\text{Choose } \delta = \min \left\{ 1, \frac{\varepsilon}{5} \right\}$$

$$|x - 2| < \frac{\varepsilon}{5}$$

$$5 |x - 2| < \varepsilon$$

$$|x + 2| |x - 2| < \varepsilon$$

$$\text{therefore } |x^2 - 4| < \varepsilon$$